

Chapter 3: Partial Derivatives

MTH 202

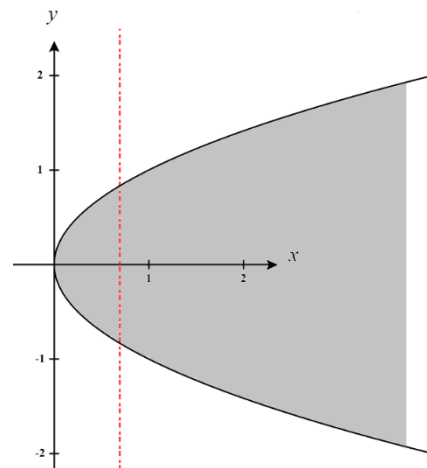
3.1 – 3.5 Review Worksheet (In-Class)

1. State the domain for the function $f(x, y) = \frac{\sqrt{x-y^2}}{e^x - 2}$ and create a picture of the domain in the xy -plane.

We see that $x \neq \ln(2)$ & $x - y^2 \geq 0$ for this function to be defined.

Therefore, we define the domain as: $D = \{(x, y) | x \neq \ln(2) \text{ \& } x \geq y^2\}$

. A picture of this domain is shown at the right (notice that we have excluded all points *on* the line $x = \ln(2)$).



2. Determine each of the following limits or show that it does not exist.

a) $\lim_{(x,y) \rightarrow (2,1)} \frac{xy}{x^2 + 3y^2}$

We note that we are computing the limit of a rational function here which is continuous at $(2,1)$,

$$\text{therefore } \lim_{(x,y) \rightarrow (2,1)} \frac{xy}{x^2 + 3y^2} = \frac{(2)(1)}{(2)^2 + 3(1)^2} = \frac{2}{7}$$

b) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + 3y^2}$

If we approach $(0,0)$ along the path $y = 0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + 3y^2} = \lim_{(x,0) \rightarrow (0,0)} \frac{x(0)}{x^2 + 3(0)^2} = \lim_{x \rightarrow 0} \frac{0}{x^2} = \lim_{x \rightarrow 0} 0 = 0$$

If we approach $(0,0)$ along the path $y = x$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + 3y^2} = \lim_{(x,x) \rightarrow (0,0)} \frac{x(x)}{x^2 + 3(x)^2} = \lim_{x \rightarrow 0} \frac{x^2}{4x^2} = \lim_{x \rightarrow 0} \frac{1}{4} = \frac{1}{4}$$

This implies that the limits across all paths are not the same and therefore the limit DNE.

c)
$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2(y^3)}{x^4 + y^{12}}$$

If we approach $(0,0)$ along the path $y=0$ we see that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2(y)}{x^4 + y^4} = \lim_{(x,0) \rightarrow (0,0)} \frac{x^2 \sin^2(0)}{x^4 + (0)^4} = \lim_{x \rightarrow 0} \frac{0}{x^4} = \lim_{x \rightarrow 0} 0 = 0$$

If we approach $(0,0)$ along the path $y = \sqrt[3]{x}$ we see that

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2(y^3)}{x^4 + y^{12}} &= \lim_{(x, \sqrt[3]{x}) \rightarrow (0,0)} \frac{x^2 \sin^2\left(\left(\sqrt[3]{x}\right)^3\right)}{x^4 + \left(\sqrt[3]{x}\right)^{12}} = \lim_{x \rightarrow 0} \frac{x^2 \sin^2(x)}{x^4 + x^4} = \lim_{x \rightarrow 0} \frac{\sin^2(x)}{2x^2} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin^2(x)}{x^2} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin(x)}{x} \right)^2 = \frac{1}{2} (1) = \frac{1}{2} \end{aligned}$$

This implies that the limits across all paths are not the same and therefore the limit DNE.

3. Suppose that a function $f(x, y)$ is continuous over all of $\mathbb{R}^2$ and has the following properties. First suppose that for any point (a, b) it is the case that $f(a, b) = -f(b, a)$. Second suppose that the following points exist on the surface created by $f(x, y)$:

$$(1, 2, 3), (2, 5, 10), (3, -4, 5), (-1, 5, 2), (4, 7, 1), (-4, -3, 1)$$

Use this information to determine the exact value of the limit $\lim_{(x,y) \rightarrow (-4,3)} f(x, y)$.

Since we are told that $f(x, y)$ is continuous over all of $\mathbb{R}^2$ we know that $\lim_{(x,y) \rightarrow (-4,3)} f(x, y) = f(-4, 3)$.

Based on the first property we are told about the function we know that $f(-4, 3) = -f(3, -4)$.

Then, based on the list of points given we know that $f(3, -4) = 5$.

Therefore, we have that $\lim_{(x,y) \rightarrow (-4,3)} f(x, y) = f(-4, 3) = -f(3, -4) = -5$

4. Compute each of the indicated partial derivatives for the function $f(x, y) = x^2 y^3 + \frac{y+x}{\sin(x)}$.

a) Find f_x .

$$f_x(x, y) = 2y^3 x + \frac{\sin(x)(1) - (y+x)\cos(x)}{\sin^2(x)} = 2y^3 x + \frac{\sin(x) - \cos(x)(x+y)}{\sin^2(x)}$$

b) Find f_{yx}

$$\text{First note that } f(x, y) = x^2 y^3 + \frac{y+x}{\sin(x)} = x^2 y^3 + \frac{y}{\sin(x)} + \frac{x}{\sin(x)}.$$

Then we calculate that

$$f_y(x, y) = 3x^2 y^2 + \frac{1}{\sin(x)} = 3x^2 y^2 + \csc(x)$$

$$f_{yx}(x, y) = 6xy^2 - \csc(x)\cot(x)$$

5. Like in the last question, consider the function $f(x, y) = x^2 y^3 + \frac{y+x}{\sin(x)}$

a) Show that the function is differentiable at the point $\left(\frac{\pi}{2}, \pi\right)$.

We note from the previous question that

$$f_x(x, y) = 2y^3 x + \frac{\sin(x) - \cos(x)(x+y)}{\sin^2(x)} \text{ and } f_y(x, y) = 3x^2 y^2 + \frac{1}{\sin(x)}.$$

We see that these partial derivatives exist and are continuous at the point $\left(\frac{\pi}{2}, \pi\right)$. Therefore, $f(x, y)$ is differentiable at $\left(\frac{\pi}{2}, \pi\right)$.

b) Find a point where we cannot guarantee that the function is differentiable.

We see that these partial derivatives do NOT exist at the point $(0, 0)$, since $\sin(0) = 0$. Therefore, $f(x, y)$ is not necessarily differentiable at $(0, 0)$.

6. Determine $\frac{\partial z}{\partial y}$ for the surface given by the equation $xz + yz^2 = \ln(y + z + x)$

We proceed by taking the partial derivative with respect to y implicitly.

$$\begin{aligned}\frac{\partial}{\partial y}[xz + yz^2] &= \frac{\partial}{\partial y}[\ln(y + z + x)] \\ x \frac{\partial z}{\partial y} + \left(y2z \frac{\partial z}{\partial y} + z^2 \right) &= \frac{1}{y + z + x} \left(1 + \frac{\partial z}{\partial y} \right) \\ x \frac{\partial z}{\partial y} + y2z \frac{\partial z}{\partial y} + z^2 &= \frac{1}{y + z + x} + \frac{1}{y + z + x} \frac{\partial z}{\partial y} \\ x \frac{\partial z}{\partial y} + y2z \frac{\partial z}{\partial y} - \frac{1}{y + z + x} \frac{\partial z}{\partial y} &= \frac{1}{y + z + x} - z^2 \\ \frac{\partial z}{\partial y} \left[x + y2z - \frac{1}{y + z + x} \right] &= \frac{1}{y + z + x} - z^2 \\ \frac{\partial z}{\partial y} &= \frac{\frac{1}{y + z + x} - z^2}{x + y2z - \frac{1}{y + z + x}}\end{aligned}$$

7. Suppose that $z = x^4 + x^2y$, $x = 2t + 5$, $y = \sin(t) - e^{2t}$. Find $\frac{dz}{dt}$ when $t = 0$.

We know that $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} = (4x^3 + 2xy)(2) + (x^2)(\cos(t) - 2e^{2t})$

Therefore,

$$\begin{aligned}\left. \frac{dz}{dt} \right|_{t=0} &= (4(5)^3 + 2(5)(-1))(2) + (5)^2(\cos(0) - 2e^0) \\ &= (500 - 10)(2) + 25(1 - 2) \\ &= 980 - 25 \\ &= 955\end{aligned}$$

8. Suppose that $z = \frac{x}{2y+x}$, $y = pq\sqrt{r}$, and $x = p\sqrt{qr}$. Find $\frac{\partial z}{\partial p}$ when $p=2$, $q=1$, $r=4$.

We know that

$$\begin{aligned}\frac{\partial z}{\partial p} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial p} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial p} \\ &= \left(\frac{(2y+x)-x}{(2y+x)^2} \right) (\sqrt{qr}) + \left(\frac{(2y+x)(0)-(2)(x)}{(2y+x)^2} \right) (q\sqrt{r}) \\ &= \left(\frac{2y}{(2y+x)^2} \right) (\sqrt{qr}) + \left(\frac{-2x}{(2y+x)^2} \right) (q\sqrt{r})\end{aligned}$$

Therefore,

$$\begin{aligned}\left. \frac{\partial z}{\partial p} \right|_{p=1, q=1, r=4} &= \left(\frac{2(4)}{(2(4)+8)^2} \right) (\sqrt{1}(4)) + \left(\frac{-2(8)}{(2(4)+8)^2} \right) (1\sqrt{4}) \\ &= \left(\frac{8}{(16)^2} \right) (4) + \left(\frac{-16}{(16)^2} \right) (2) \\ &= \frac{2}{16} - \frac{2}{16} \\ &= 0\end{aligned}$$

9. A manufacturer has modeled its yearly production function P (the value of its entire production in millions of dollars) by the function $P(L, K) = 1.47L^{0.65}K^{0.35}$. L represents the number of labor hours (in thousands) and K represents the invested capital (in millions of dollars). Suppose that when $L = 30$ and $K = 8$, the labor force is decreasing at a rate of 2000 labor hours per year and the capital is increasing at a rate of \$500,000 per year. Find the rate of change of production. Round your final answer to 3 decimal places.

We want to find $\frac{dP}{dt}$ and from the chain rule we know that $\frac{dP}{dt} = \frac{\partial P}{\partial L} \cdot \frac{dL}{dt} + \frac{\partial P}{\partial K} \cdot \frac{dK}{dt}$.

Therefore, we know that

$$\begin{aligned}\frac{dP}{dt} &= 1.47(0.65)L^{-0.35}K^{0.35} \cdot (-2) + 1.47(0.35)L^{0.65}K^{-0.35} \cdot (0.5) \\ \left. \frac{dP}{dt} \right|_{L=30 \text{ \& } K=8} &= 1.47(0.65)(30)^{-0.35}(8)^{0.35} \cdot (-2) + 1.47(0.35)(30)^{0.65}(8)^{-0.35} \cdot (0.5) \approx -0.596 \\ \left. \frac{dP}{dt} \right|_{L=30 \text{ \& } K=8} &\approx \$596,000 / \text{year}\end{aligned}$$

